Gavin Onghai

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Education

Yale University New Haven, CT

B.S. Electrical Engineering and Computer Science (GPA 3.80/4.00) *Expected Graduation May 2028*

- **Hahn Scholarship**: 6 selected from 57,517, **Alan S. Tetelman Fellowship**, **Yale CS Department Keidel Grant**
- **Relevant Coursework:** Data Structures, Algorithms, Systems, Distributed Systems, Database Systems, Probability
- **Activities:** Yale IEEE - President, Yale Computer Society - Yale Meals Team Lead, Yale Quant Finance Organization

Experience

Machine Learning Researcher | *Python, PyTorch, OpenCV, Linux (GPU Cluster)* May 2026 – Aug 2026

Hong Kong University of Science and Technology (HKUST) *Sai Kung District, HK*

- Built the **crowd-counting** data component for a **real-time Internet of Things (IoT) distributed system**, informing a road traffic decision engine managing traffic light signaling following **50,000-attendee** events.
- Cut crowd-counting model error by **23.9%** MAE (1.80 to 1.37) versus a single-frame baseline by encoding **temporal convolution** and **people motion** over 3-frame windows into the **convolutional neural network** using **PyTorch**.
- Automated human motion data labeling across **14,000 video clips** using Farneback optical flow in OpenCV.

Software Engineer Intern | *Python, FastAPI, PostgreSQL, Docker, React* Aug 2025 – Dec 2025

NIRA AI (withnira.com) — South Park Commons Backed AI Startup *Sunnyvale, CA*

- Built and scaled a **large language model (LLM)** professional networking agent from scratch to **1,000+ daily users**.
- Cut response latency by **75%** (**20s to 5s**) by designing **RESTful APIs** in **FastAPI** with batched prompting.
- Architected multi-source **ETL data pipelines** (Gmail, Google Calendar, LinkedIn), processing **3 million** emails into a **PostgreSQL/pgvector** store to power the **agentic advisor** and **semantic search** across LinkedIn user profiles.
- Built an **asynchronous webhook service** that runs **LLM calls** on emails to detect positive introductions and prompt users to send gifts using the **Stripe API**, processing **500+ leads** and creating a second monetization channel.
- Built end-to-end **monitoring and alerting** via **Site24x7** with **Slack** integration; caught **5** critical errors before they reached users and improved system **uptime** to **99.9%**; incorporated feedback from the founders to prioritize fixes.

Undergraduate Researcher | *Python, C, NumPy, Pandas, scikit-learn* Apr 2025 – Aug 2025

Yale School of Engineering and Applied Science *New Haven, CT*

- Built a low-cost lensless camera on a **Raspberry Pi** that measures polarization in a single shot.
- Implemented an **object-oriented** high-dimensional inverse solver in **Python** for image reconstruction, and trained a **scikit-learn regression** model mapping RGB values to polarization, achieving **4.68%** single-shot polarization error.
- Automated data acquisition with **serial-port scripting in C**, processing **100,000** RGB frames in **NumPy/Pandas**.

Projects

Jera: Satellite Alt-Data Signal Pipeline | *Python, Sentinel-1 SAR, FastAPI, PostgreSQL, R2, CI/CD* May 2026

- Built a data pipeline ingesting satellite images to compute a mine-activity score for equity forecasting; shipped a **FastAPI/PostgreSQL** backend with point-in-time queries, idempotent ingest, and **Parquet** exports to **R2**.

Recruitimation: Staffing Automation for IT-RS.net | *TypeScript, Fastify, PostgreSQL, Railway* Jun 2026

- **Solo-built** and **deployed** an **async** candidate-screening system for a staffing agency: a **BullMQ/Redis** worker ingests LinkedIn profiles, ranks them using **OpenAI embeddings**, and stores results in **PostgreSQL** (Prisma) for review.

Systems Programming Projects | *C, C++, BASIC* Mar 2026

- **One-Pass Transpiler:** Engineered a recursive-descent LL(1) **BASIC** to **C** transpiler in **C++**; optimized performance by bypassing AST overhead for direct one-pass code generation using a symbol table.
- **Unix Shell:** Architected a systems-level **IPC engine** managing **multi-stage process execution** via signal handling and file-descriptor manipulation; supports **asynchronous execution** and logical chaining.

SmartBell: Real-Time Embedded CV Doorbell | *C++, Python, TensorFlow Lite, InsightFace* May 2026

- Built an **edge CV pipeline** on Raspberry Pi 4: **C++** process talks over a Unix socket to a **Python** server running **quantized MobileNet-SSD** (TFLite) and **RetinaFace+ArcFace** face recognition; achieved **300ms** E2E latency.

Technical Skills

Languages: C, C++, Java, Python, x86-64 Assembly, SQL, TypeScript, JavaScript, Go
AI/ML: LLMs, prompt engineering, embeddings, semantic search, vector databases (pgvector), model evaluation
Backend / Web: REST APIs, ETL pipelines, FastAPI, Node.js, PostgreSQL, Redis, React, Next.js, microservices, Fastify
Tools: gdb, Valgrind, Git Version Control, Docker, Linux (Daily User), Bash, CI/CD (Github Actions), AI Coding Tools
Libraries: NumPy, Pandas, scikit-learn, PyTorch, OpenCV, TensorFlow Lite